Systems Engineering Unit 1&2 Design Brief

Systems Engineering Unit 1&2 Design Brief
William Emery, 23/02/2026, St. Helena Secondary College

Folio sections covered in tab: 1.1, 1.2, 4, 6

Design overview & brief (1.1, 1.2)

In 2026, I will be designing and producing my project, OpenOrgel. The project itself is a portative pipe organ of which the documentation, design, and software will be open source. It will be mainly constructed of wood and 3D Printed PLA (Polylactic Acid) plastic. The organ will be powered by a self designed electric centrifugal blower that runs off a battery or voltage limited mains less than 50v AC or 120V ripple-free DC , preferably less than 50V to power a DC motor that will run it. It will have a closed-loop wind system allowing for precise control over dynamics and air pressure. The wind system will make use of sensors to monitor air pressure, which will be generated passively via a weighted bellow lid on the reservoir/regulator. There will also be a tachometer or hall sensor to measure the RPM of the impeller, which will be reported to the controller. If air pressure drops, the system will increase the RPM accordingly so as to not overcompensate. If air pressure rises, the system will decrease the RPM, or if incredibly extreme the system will open some relief valves. To keep pressure stable past the regulator, there will be a one way valve to prevent backflow, and before that a valve that only lets air in at a certain pressure. This pressure will have to be adjustable by the controller in response to user input from the dynamics console.
The current, most complicated form will have a range of C0-C8 which is 8 octaves of full chromatic range, larger than the piano. To allow for portability and size constraints, the low (pedal) range, which will be from C0-F#3 will be synthesised. Other voices may also be synthesised throughout the entire range of the organ, such as *Voix Celeste* or reed stops, or 8' up to 32' stops that are simply too large to produce or difficult to manufacture. I may add a percussion console which has some percussion instruments (Bells, drums, etc) that will likely be synthesised, as the large pipe organs of today typically have that. I will also make a synthesised *Digeridoo* stop that sounds exactly like an indigenous Digeridoo, perhaps with more indigenous instruments. The indigenous instruments will be implemented ethically, considering the context surrounding the instruments and the implications of including them, and will be approached with utmost respect.

1.2)
The instrument is designed to function as an accessible, affordable, consistent, easy to use, easy to repair instrument for educational purposes

In modern society, there is a persistent imbalance of the standard of living between nations. Developed nations enjoy a much more comfortable standard of living with prevalent access to resources such as musical education, which in the developed world is considered a right. In developing/underdeveloped nations, the standard of living is considerably less luxurious compared to developed nations. In these places, musical education is difficult to come by due to the intrinsic nature of musical education to be expensive. Musical education that is provided is typically not comprehensive and not to an acceptable standard, compared to more economically developed nations. Many across the world believe that musical education is an important issue in childhood, as it is necessary for the preservation of cultural music and indeed necessary for the preservation of culture itself. Unfortunately, according to The Age (The Age), The Music Trust Fund found that 63% of responding schools offered no music instruction, and only 23% of government school music programs were taught by specialist music teachers. For context, Australia is regarded as one of the richest countries in the world, consistently ranking second in median wealth per adult behind Luxembourg.

The pipe organ is widely regarded as one of the most versatile instruments out there, due to the variety of sounds, timbres, usage of standard keyboard format, and pure range it possesses. For this reason, the pipe organ would function as a suitable musical instrument for musical education, and thus should be used for education. Unfortunately, this "king of instruments" is dying out due to the cost, importability, mechanical complexity, and maintenance costs associated with the instrument. An average pipe organ for a church would in this day and age cost between $200,000 and several million dollars. According to Yamaha, it can take up to five years from proposal to completion. The instrument itself is so large that the churches it is typically held in are generally built/designed around the pipe organ. The instrument is very mechanically complex, having hundreds of moving parts, springs and mechanical linkages all interconnected to move a single rank of pipes. This adds to the intense cost of the instrument, which is further amplified by the sheer number of pipes used in these instruments. An average non-portatative pipe organ will likely have more than a thousand separate pipes. The traditional large pipe organ is also by design non-portable, a fixed installation, as it is built into buildings and tedious to remove, and not designed to run free-standing. These factors combine to clearly demonstrate that access to musical education through an instrument as versatile as the pipe organ is incredibly challenging to come across for people without the funds or reliable access to buildings housing pipe organs. This could prove damaging to the preservation of the history of music as the pipe organ has been used throughout western music since the ancient Greek "Hydraulis". The OpenOrgel aims to create a full range, portable and disassembleable pipe organ which will be fully open source, and combines mechanical action, digital action, acoustic sounding pipes and digitally synthesised sounds. The OpenOrgel is aimed to preserve western and indigenous Australian culture in music and allow for cheaper yet more extensive musical education in countries where musical education is scarce, or poverty is rampant, and furthermore to solve the issue of importability, size, and inaccessibility surrounding the pipe organ.

Design Criteria (6, 3.2)

Functional requirements:

- Provide an acoustic sounding range spanning a minimum range of 2 octaves.
- Provide a digitally synthesised sounding range across a minimum range of 4 octaves.

- Provide voicing options across synthesised and digital range.
- Utilise a closed-loop wind system which takes feedback from sensors to adjust air pressure and blower RPM, with air pressure stable within 5% of 85mmWC .
- Produce a mechanical tracker action for acoustic range
- Allow for MIDI programmability across synthesised range
- Produce a percussion extension module
- Operate within 50V AC or 120V Ripple free DC.

Non functional requirements:

- Maximum size: Approx. width and length of tables in T2, no taller than waist height (Reference frame for design, not final!)
- Must make minimal use of non-renewable or non-biodegradable resources (Measurement/goal to be added)
- Must minimise environmental impact (Kg of co2 released by production of individual systems)
- Durability- Must survive multiple (more than 20) cycles of deconstruction and reconstruction.
- Must keep wind pressure stable within 5% of selected air pressure (mmWC)
- Must be relatively simple to maintain (with comprehensive guides and documentation), tracker action should be durable (see durability) and easy to replace should it become damaged (taking no longer than 10 minutes to replace an entire key's tracker action system)
- Must be relatively simple to disassemble- no permanent joins aside from components of (sub)systems. (No longer than 30 minutes to disassemble into its most compact form)
- <5ms latency for digitised stops at all times, regardless of load (unless a power surge)
- Blower noise must not exceed ~65 dB measured at 1 metre from the console (manuals) of the instrument, under normal playing conditions and exceptional conditions (See definitions), for 30 seconds at nominal air pressure. Longer operation with exceptional conditions is likely to cause temporary pressure instability, or motor shutdown to protect valuable components. Blower noise should not exceed ~68 dB measured at 1 metre from the console under extreme conditions at nominal air pressure for 10 seconds. Extension past this will likely result in air pressure instability, damage to the wind system, failure to sound, or motor shutdown.
- Digital systems should ultimately be compatible as a MIDI controller for programs like Hauptwerk
- Comprehensive documentation and technical instructions for maintenance of all components of the instrument

Definitions:
Normal playing conditions- Nominal air pressure, 2 acoustic stops open, C7 chord being played (4 notes at the same time).
Exceptional playing conditions- Nominal air pressure, 3-4 acoustic stops open, any 6 notes played at the same time.
Extreme playing conditions- Elevated (88mmWC) air pressure, 4+ acoustic stops open, any 8 notes played at the same time

Subsystems:
- Wind system
  - Blower
  - Regulator
  - Tubing
  - Windchest
- Action
  - Keyboards
  - Tracker action
  - Valve pallets
  - Mechanical stops
  - Digital keyboards
  - Digital stops
  - Tracker action sensing
- Acoustic pipes
- Digitised stops
- Control electronics
- UI
  - Programme selector screen
  - Wind regulation/info screen
  - Dynamics selection
  - Modular Expansion System Interface (MESI, allows for connection of expansion modules like percussion)
- Audio output (Amp, Speakers)

Success criteria:
- Latency of under 5ms with digital stops.
- Fitting size constraints.
- Wind regulation within 5% (mmWC) at all times except when changing between dynamics or powering on/off.
- Voltage stability within 5% of desired nominal voltage.

- Disassembly within 30 minutes.

Minimum Viable OpenOrgel specifications:

- At least 4 octaves of total sounding range, 1.5 octaves minimum acoustic range, minimum of 1 voicing acoustic stop, 2 synthesised stops, 1 indigenous instrumental stop, synthesised or otherwise.
- Functionally identical wind system aside from size, same specifications for tolerances.
- MESI interface with percussion module synthesised or otherwise

Product expected to be completed:

- Approximately 6 octaves of sounding range (C2-C8), 2 octaves of acoustic range (C5-C8) and 6 octaves digital range.
- 1 Voicing acoustic stop
- 2-4 synthesised stops
- 1 indigenous instrumental stop (Synthesised)
- No pedalboard
- 6 octave manual
- MESI interface with simple percussion module, synthesised
- Functionally identical wind system, designed to work for the chosen 2 octaves of acoustic range.

Constraints:

- Size must NOT exceed 120cm in total length, 60cm width, and 60cm depth, not including blower, regulator, MESI, or stop console.
- Cost must not exceed 400 AUD. (Unfinalised, subject to change during production. Not including supplies or materials already owned or supplied by school.)
- Must be completed in documentation and production by 2 weeks into term 4, TBC.
- Must make use of Mechanical and Electrotechnological systems and effectively combine them.
- For non electrical parts, materials available to be used:
    - Plywood
    - Timber
    - MDF
    - Acrylic
    - 3D Printed PLA plastic
    - 3D Printed PETG plastic

    - Machined metal (TBC)
    - Sheet metal (TBC)
    - Metal Piping
    - PVC Piping
    - Flexible plastic piping
- Safety
    - No unmitigated or undeclared potential to harm or maim life.
    - Fire risk must be effectively managed
    - No feasible potential for death caused by failure of safety mechanisms and design.
    - Must comply with electrical regulations- Not exceeding 50V AC or 120V Ripple free DC.
    - Compliant with AS/NZS 3100:2022
    - Compliant with AS/NZS 3760:2022

Stakeholders

There are 3 main stakeholders. The user, the wider open source community, and people who own the instrument (if not the user). The user requires the pipe organ to be functional and consistent in its sound, intuitive user interface, and that the motor noise does not overpower the noise of the pipes or synthesiser. The user further requires a smooth mechanism and seamless transition between acoustic and synthesised notes, as well as a wide selection of synthesised sounds across a wide range of octaves The wider open source community requires that the product be entirely open source, with accessible and comprehensive documentation surrounding its assembly, design, usage, maintenance and expansion. The owners call for the product to be portable, easy to maintain, and inexpensive, as well as being easy to assemble and use, with comprehensive documentation.

Testing

The scale of this project calls for immense testing to be completed before any finalised product may be completed. In this document I will have a list of tests to be performed, their results, why the test is to/has been performed, why the results matter, how the results will be used, and their implementation. There will also be photographic records and data records uploaded. The tests **are not** in order of completion.

A large amount of Test Data, namely code, is to be found at my github repository. This is linked here. It also contains all my 3d files and every test file as of yet. Photographs are to be uploaded to the document first, and then github if deemed needed.
https://github.com/williamemery2010-debug/OpenOrgel/tree/main#

Test no.	Reason for test	Overview of test	Method	Results
1.	To determine if the motor is actually able to handle the RPM needed under load.	A test of the motor and gearbox. The motor and gearbox are to be connected. The motor is to be	1: Power on motor, and stand a safe distance away. 2: Observe the motor running between	

		connected to an arduino and power supply. The shaft will be connected to a PETG impeller, and a tachometer. RPM and peak current as well as temperature , and noise level will be measured.	stages of testing. Observe RPM. 3: When code reaches an extreme stage, allow the motor to run for 50 seconds. 4: Stop testing. 5: Collate data from sensors over runtime into a spreadsheet. 6: Graph data.	
2.	To determine if the regulator is suitable and able to maintain pressure.			
3. Impeller scale testing	To determine impeller pitch for usage in the			

	wind system. To be done at small scale.			
4. Synthesiser testing.	To determine if the synthesiser program is fit for use in OpenOrgel.	The synthesiser will play a test MIDI file, namely Bach's Fugue In G Minor - 'The Little'. I will listen to this file and if the program produces sounds which sufficiently please my ears, I will declare it fit. This is done for every stop.	Open Synthesiser. Load MIDI file. Select stops. Play file. Adjust harmonics until satisfied. Repeat.	Current version as of 03/06/2026 is acceptable.

Social, Environmental and Ethical impacts (4)

The OpenOrgel project aims to have minimal negative social, environmental or ethical impacts. In order to achieve this, the OpenOrgel will only use natural, reusable, renewable, recycled, or biodegradable materials where possible, such as Aluminium sheet metal (if used) or MDF, plywood, or even PLA plastic, which is biodegradable (in certain conditions), renewable, and recyclable. I will avoid using materials that do not possess any of the properties stated earlier, and if possible I will not use them whatsoever, except in electronics. The impact of the use of all these materials is still however present, cutting down trees for MDF and plywood is damaging to environments as it removes natural habitats for animals like birds or critters. Therefore, I will aim to use sustainably sourced MDF and plywood. For metals, the process of production uses intense amounts of energy, which is typically sourced from fossil fuels. Unfortunately, as a consumer, I have very little control over this. Thus, I will simply avoid metalwork wherever possible. There is also an environmental impact from the use of 3D printers which uses a lot of energy, which is also sourced from mainly fossil fuels. To curb this, I will mainly be using my home 3D printer where possible, which is mostly powered by our solar panels and solar battery. There is also environmental impact from the production of PLA plastic, releasing harmful chemicals, yet there is not much I can do to curb this. If possible I will source recycled PLA plastic to combat waste. For social impacts, the only impact OpenOrgel aims to make is making musical education more accessible for all levels of musical education, and to generally increase the amount of musical education in the world. A side effect of this would be a heightened preservation of the musical arts, as more people would know how to play the music and keep it alive, and thus help to preserve many cultures around the world which may be at risk due to poor or minimal musical education.

Systems thinking (3)

Here is the system block diagram.

Here is the IPO Diagram:

Input	Process	Output
240v AC	Transformed to 120VDC	Heat and 120VDC
On switch	Blower spins up	Heat and Sound

Organ Stop Console Switches	Arduino/Teensy reads change, synthesiser applies new registrations	New stop waveform is added to synthesis pool, sounds when keys depressed
Dynamics selections	Microcontroller reads change, synthesiser changes volume of speakers, PWM is slightly altered to change air pressure, load on bellow lid is slightly changed to alter air pressure	Volume decreases or increases
Air pressure reading	Microcontroller reads change, processes and determines if in nominal range, if not, PWM of motor is altered, if needed load on bellow lid is changed, if needed emergency pressure relief valves open in reservoir and bellow.	Air pressure is returned to nominal conditions.
Air sucked into blower	Air is directed into bellow and reservoir, pressured, and put into windchest.	When pallet valves open, air enters pipe and sounds the note.
Key position changes	A digital switch is triggered, tracker action is mechanically moved, valve pallets open, microcontroller recognises change in switch state and sends a signal to the computer, computer determines note as MIDI input and relays to synthesis program, synthesis program synthesises note, sends note state to speaker.	Note sounds through speakers and/or pipes.

Safety & Risk management (9)

As with all projects, there are risks associated with OpenOrgel.
Low priority- Prior 1-4.
Medium priority- prior 5-9.
High priority- Prior 10-12.
Urgent Action- Prior 15+.

Risks:

- Impeller material failure due to excessive force from revolving too quickly for the used material. Can be combatted with software limitations- if the motor reaches an unsafe speed, the controller will reduce RPM and (if implemented) open reservoirs. Unlikely- 2. Minor injury- 2. Prior 4. Assessment: Test blower impeller limitations to determine maximum safe operating speed.
- Fire risk- PLA is a flammable plastic, and if used to house the motor for the blower may deform, melt, or burn from the heat generated by the motor. Could be combatted with noctua style fans to quietly cool the motor. I may manufacture a heatsink from machined aluminium for the motor as well. Unlikely, 2. Minor Injury, 2. Prior 4. Assessment: Implement cooling measures.
- Explosive failure in blower: If using 3D printed plastic for the impeller, the impeller blades are likely to delaminate and fling shards at incredibly high speeds, which could prove damaging to life. The housing of the impeller could also deform from the heat of the motor, and perhaps collapse into the impeller, causing an explosive failure, flinging plastic shards at high speeds once more. Highly unlikely, 1. Minor injury, 2. Prior 2. Assessment: No action needed at this time.

Project related non-harmful risks.

- Wind oscillation from incorrect PID tuning- To be avoided by extensive simulation using softwares like Fusion 360 or Autodesk CFG, and extensive practical testing.
- Excessive blower noise leading to poor sound quality- could be avoided by enclosing the blower in a soundproofed chamber. Keep a foam block in between the blower and ground to prevent excessive noise from ground vibrations.
- Mechanical wear in tracker action- Use durable materials. Test from a selection of materials to determine the best material. Test for durability over 100 key depressions, compared to weight and size.
- Latency for digitised systems exceeding ~5ms under processing load- Test a variety of controllers. Find which one handles processing best under intense load. Install a heatsink and cooling fans on the controller to avoid thermal throttling.
- Structural instability after repeated assembly/disassembly- Test. Edit design accordingly.
- Project scope exceeding available time- At first build the MVP, taking into account how long each system takes to create at that scale. Expand the systems to their maximum possible scale with the given time frame. Calculate pipe proportions beforehand, ensure the blower system can theoretically handle the highest spec form.

Work practices for risk mitigation:

In order to minimise the risks dealt with during the production and testing of this project, I will have to create some safe work practices.

1. When testing a full sounding system, if the noise level exceeds 70 decibels I will use hearing protection.
2. When using power tools, I will follow the posted safe work procedures and use appropriate PPE.
3. When conducting practical work such as woodwork, I will ensure to use eye protection and closed toe leather shoes.
4. I will attempt to insulate all conductive material in the project to minimise electrocution risk.
5. Where insulation is not possible for practical reasons, I will use gloves and I will ground myself. I will also ensure to turn off all electrical systems and remove them from power before working on the electrical components.
6. Follow all safe work procedures.
7. Read MSDS of all materials used. Upload to here.

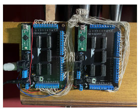
Work Cited

The Age. “Learning music educates mind and soul, but often falls to school budget cuts.” *The Age*, 4 May 2015, https://www.theage.com.au/education/learning-music-educates-mind-and-soul-but-often-falls-to-school-budget-cuts-20150504-1na67a.html. Accessed 1 03 2026.

Research

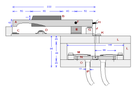
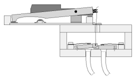
Research

No.	Pic	Relevance	Source
1		An incredibly compact design, helpful for design of the PVC piping. It uses tubing though, and I want to avoid this. It also doesn't have too much range.	https://www.instructables.com/PVC-Pipe-organ/ 1031 hrs 9/02/2026

2		A very compact design, especially considering its range. I will be referencing this design for the mechanisms for the keys and the blower, and potentially pipe design. It has a decently large low end which I would like to take inspiration from. I do not like how it isnt very portable, but it can be disassembled for transport. A very good source for the blower design- it has suggested a vacuum motor. This refers to the size constraint.	https://www.sentex.ca/~mwandel/organ/organ.html 1043 hrs 9/02/2026
3:		In essence, just functional inspiration for the computer control aspect of the organ. Validates my idea to use solenoids. Also functional as a proof-of-concept that controlling a pipe organ of sorts via MIDI protocol is viable and doable.	https://www.pjrc.com/church-organ-midi-control/ 0959 Hrs 10/02/2026

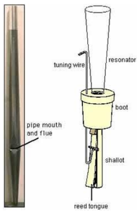
4.	No helpful images.	This is the most detailed method for a fully computer controlled pipe organ, aside from a physical keyboard (AKA A manual.). If I were to consider a fully MIDI controlled organ I would extensively use this, however I am constrained by a necessity to use mechanical systems. This is still helpful for the programming side of things. There is still a need for mechanical systems with the solenoid actuation of valve pallets. This refers to the constraint of using mechanical systems.	https://www.instructables.com/Adding-MIDI-to-Old-Home-Organs/ 1003 Hrs 10/02/2026
5.		A relatively inexpensive solenoid that will do what I need it to do (pull valve pallets) and function as a spring. At time of writing it costs 19.95 which for 4 octaves of full chromatic range for 1 rank would cost… 960 dollars. No way. This refers to the cost restraint.	https://www.amazon.com.au/HALLMERS-Actuators-Intermittent-Electromagnet-Industrial/dp/B0CZR8PKT2 1015 Hrs 10/02/2026

6.	The image refused to paste, but just imagine it to be a regular like aluminium shell DC motor.	Much cheaper than solenoids at $9.49 for a pack of 6 motors. Should be strong enough to move the valve pallets. About $75.92 for 4 octaves of one rank. Much more economically viable and actually achievable with my financial situation, would require a redesign of my system. I will have to use cables to actuate the valve pallets and gears to turn and coil the cable from the key OR the motor. I will make some diagrams. Cost restraint as well.	https://www.amazon.com.au/1-5-6V-15000-16500RPM-Electric-Science-Experiments/dp/B073Q2Y3RC/ref=asc_df_B073Q2Y3RC?mcid=bd5a29796a5c365c8e57b334217e289d&tag=googleshopdsk-22&linkCode=df0&hvadid=712243389570&hvpos=&hvnetw=g&hvrand=6409822242168328328&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9071368&hvtargid=pla-447885543765&psc=1&gad_source=1 10125 Hrs 10/02/2026

7.	 	This is likely the mechanism I will use. I have come to realised that MIDI control will be incredibly difficult to implement with DC motors as it would require the use of gears to pull down the key or otherwise pull the valve pallet up, which I cannot at this time figure out how to do. Solenoids would make it much easier, but they are absurdly expensive. I could maybe use servos to move the keys, or even to move the valve pallet. I will not use DC motors to implement MIDI control if implemented at all. An idea I had was to use potentiometers on the valve pallets to measure how open they are, and relay that to other stops with different voiced pipes so there is no need for mechanical stops. I could also attach a contact to the bottom of the key and what I will call the key stop, which stops the key from travelling too far, which would do the same job as the potentiometers. This would mean I would forfeit midi control for optional voicings. I would ensure that the other stops are only sounded if they are enabled via a button. This refers to the constraint of cost, size, mechanical and electrotechnological systems, and timeframe.	1219 Hrs 12/02/2026 See no. 1 for link

8.		Wooden pipes will make up the majority of the pipes in my organ. This is because they are easy to design and manufacture, and inexpensive. I will use the 3d printers however to print the blocks of the pipes, which have complicated internal geometry which would be difficult to manufacture of wood. Another good thing about the use of wooden pipes is that they are quite obviously biodegradable and renewable, the whole pipe, including the block will be biodegradable if I use Polylactic Acid (PLA) filament. This refers to the constraints of budget and timeframe.	https://www.yamaha.com/en/musical_instrument_guide/pipeorgan/mechanism/mechanism003.html 12:29 Hours 12/02/2026

9.		I was considering using PVC pipes for the lower register as it is cheap and easy to access, but it is not very sustainable and does not sound or look very good, being a bit tinny and empty. The fipple geometry would also be tricky to make with the PVC, and the pipes would be quite fragile. I am going to avoid using PVC pipes as much as possible.	https://www.instructables.com/PVC-Pipe-organ/ 12:32 Hours 12/02/2026
10.		I very briefly considered the use of metal pipes but decided against using them as they would require casting of metal, which would be incredibly expensive, defeating one of the aims of the project. This refers to timeframe and budget.	See 8 for link. 12/02/2026

11.		I considered using reed pipes to make a variety of sound and reduce overall size, but they are incredibly complex and rather annoying to manufacture and tune, so I would overall rather use flue pipes. This refers to timeframe and budget, but also mainly a personal choice.	https://www.organ.byu.edu/organtutor/organ_registration/tone_families/pipe_categories/page1.htm 12/02/2026

12.	No Image-design choice	After heavy consideration I have reached the decision to not continue with a digitally controllable fully acoustic mechanism. This is because the added complexity from the motors and changes in mechanism design would make the project entirely unrealistic for myself to achieve by the end of the year. The usage of a fully acoustic mechanism is also unrealistic and indeed a poor idea as I am required to keep a space constraint in mind- across full range the instrument will be much too large to feasibly construct in a project at the school scale, nevermind with the use of more than one sounding acoustic stop. I have decided to include acoustic stops but remove digitally controllable physical mechanisms aside from the wind system. I have also decided to reduce acoustic range to a minimum of two and a half octaves. This is making the project much more realistic in the timeframe I have. To compensate for the diminished range the system will have a set of digital sounding stops across the range. These will allow me to have full digital range that overlaps with acoustic range. The acoustic stops will have a sort of sensing mechanism using switches that detects when the tracker action mechanism is in a state that would sound a note. If the digital stops are enabled, the note will speak when the key is depressed. For keys that are not connected to acoustic stops, there will be a simplified hammer mechanism that depresses a push-button, or connects a contact, more similar to that of a piano.	3/03/2026

13:	Image refused to paste.	The decision on what switches to use on the tracker action for the digitised stops seems rather obvious to me, as a reed switch seems to be the only sensible option. A pushbutton would not function as the tracker action does not produce a vertical force in that manner to depress a push-button. A mercury switch would not work as any motion in the keys whatsoever would trigger it, but the key needs to be depressed to a certain point to trigger the sound. A rotary switch theoretically could be used, as it could provide a hinge point for certain components of the tracker action system, however they would be more difficult to physically attach to the walls of the tracker mechanism. These reed switches from aliexpress are relatively inexpensive and look decent- According to the seller they have a response time of 1ms MAX which does not exceed my <5ms latency for digitised stops limit. They also have a release time of 0.4ms maximum, which is good for ensuring that notes stop cleanly and don't 'linger'. If I run the switch at 5v and 10mA the switch is to last more than long enough.	https://www.aliexpress.com/p/tesla-landing/index.html?scenario=c_ppc_item_bridge&productId=1005007790992180&_immersiveMode=true&withMainCard=true&src=google&aff_platform=true&isdl=y&src=google&albch=shopping&acnt=742-864-1166&isdl=y&slnk=&plac=&mtctp=&albbt=Google_7_shopping&aff_platform=google&aff_short_key=UneMJZVf&gclsrc=aw.ds&&albagn=888888&&ds_e_adid=&ds_e_matchtype=&ds_e_device=c&ds_e_network=x&ds_e_product_group_id=&ds_e_product_id=en1005007790992180&ds_e_product_merchant_id=106630103&ds_e_product_country=AU&ds_e_product_language=en&ds_e_product_channel=online&ds_e_product_store_id=&ds_url_v=2&albcp=2181946380

			8&albag=&isSmbAutoCall=false&needSmbHouyi=false&gad_source=4&gad_campaignid=21819486122&gbraid=0AAAAA99aYpdQTx4DWP_MoadT29fPVFIou&gclid=CjwKCAiAh5XNBhAAEiwA_Bu8FTd2m_lkrFYq8B5xY7t_NAw70_ldafehE1jxJisyIx1e4qsBqkBflRoCnKEQAvD_BwE 3/03/2026

14:	Material choices for blower impeller, not products	My impeller will be put under intense mechanical stress at the RPM required for the airflow necessary to produce a sound. The current impeller specifications dictate that the impeller will have a diameter of 256mm, at an RPM between 6000 and 15000, which is rather fast. The formula for the stress that will be applied on the impeller centrifugally (keep in mind this is a rough estimate) is $\sigma=\rho \cdot \omega 2 \cdot r2$, Where: • σ = hoop stress (Pa) • ρ = density of material • ω = angular velocity (rad/s) • r = radius (m) = 0.128 m (half of 256 mm) If σ exceeds the tensile strength of the material, you will experience a centrifugal failure destroying the impeller and likely hurting someone. PLA plastic when 3D printed is simply not strong enough to withstand the centrifugal force of the rotation without an explosive failure, delamination, heavy deformation or centrifugal failure. The Coefficient of friction ranges from 0.5243 to 0.6702. PETG plastic is more flexible, yet somehow more stiff, and less prone to delamination than PLA, making it overall stronger. When printed, the estimated maximum safe RPM is around 14,800. This should allow for the maximal air volume. PETG also has a lower COF around 0.20-0.35, making it much more aerodynamically efficient as there is overall less drag. The flexibility of PETG means that if there is a failure, it is much less likely to fling shards at people or hurt them, and would instead deform. This is much safer for people, and in my experience people matter more than the objective.	No source, only consideration and material properties

15.		This is the brand of filament and roll I have decided to use for the impeller, should I print it. I personally trust Jaycar to provide high quality filament, and this is standard PETG that fits my standards. The maximum printing speed is 250mm/s which is well above what is needed.	https://www.jaycar.com.au/slic3d-petg-filament-black-1-75mm-1kg/p/TL6446?gad_source=1&gad_campaignid=22838588280&gbraid=0AAAAAD0dvLYZnYZtXHjkMVTDxMLlpQUCO&gclid=CjwKCAiAqprNBhB6EiwAMe3yhoK4c4wJfl20WTWMuRuIcrpBudRGSXGl7KTASuzUtDMLAcFqPtcW7hoCn2EQAvD_BwE Accessed 4/03/2026

16.		These are the motors I am considering using. The motor requires exceptionally high torque to be able to spin the impeller at the required RPM to provide the intense (upwards of 600L/min) airflow required for even a small acoustic range. The first motor I have sourced is rather borderline, it will only produce just enough torque at nominal conditions. (Operating Voltage: 12VDC At Stall: Torque: 6.5kg.cm Current: 63A Max RPM without load is 10,200RPM)	https://www.jaycar.com.au/standard-high-power-d-c-motors-9200-rpm/p/YM2776 4/03/2026 Chosen motor: T-MOTOR F50 2150KV/2200KV 6S 5inch Brushless Motor 5mm Shaft for RC FPV Freestyle Drones DIY Parts - AliExpress 26 10/03/2026

Rated Voltage (Lipo)	24V	Peak Current	54.8A
Max Thrust	2136g	Weight (excl. cable)	28.56g

KV	2200	Idle Current(10V)	≤1.6A
Max Power	1315W	Internal Resistance	47mΩ
Rated Voltage (Lipo)	24V	Peak Current	56.63A
Max Thrust	2134g	Weight (excl. cable)	28.36g

Recommended

Motor	F50	Frame Type	X4
Prop	5-inch 3-blade prop with small and medium pitch (TMHOBBY F5146)	Lipo Cells	6S
ESC	TMHOBBY F60A 8S 4IN1	Recommended takeoff weight	Within 500g
FC	TMHOBBY MINI F7 MPU6000	Max Takeoff Weight	800g

Recommended ESC settings:
AM32: start power 100%, maximum timing angle, pwm frequency ≥ 32khz, kv value slightly higher than motor kv
BL32: start power about 40%, maximum timing angle, pwm frequency: 48-byrpm
Disclaimer: Desyncs is related to many factors, including FC parameters, motor usage, installation status, etc.

Test Data

Type	Propeller	Voltage (V)	Throttle (%)	Current (A)	RPM	Thrust (g)	Power (W)	Efficiency (g/W)	Torque (N*m)	Operating Temperature (℃)
F50 2150KV	TM-HOBBY F5146	24.04	10%	0.40	5436	38	10	3.97	0.00	102
		24.02	20%	1.41	10238	152	34	4.50	0.02	
		23.99	30%	3.47	14754	341	83	4.10	0.03	
		23.95	40%	6.67	18746	572	160	3.58	0.06	
		23.88	50%	10.97	22194	824	262	3.15	0.08	
		23.79	60%	17.23	25601	1120	410	2.73	0.11	
		23.68	70%	24.76	28392	1407	586	2.40	0.14	
		23.56	80%	33.12	31026	1704	780	2.19	0.17	
		23.42	90%	42.80	33213	1975	1002	1.97	0.20	
		23.28	100%	52.52	35042	2136	1222	1.75	0.22	

Note: Motor temperature is motor surface temperature running at 80% throttle for 10 seconds.
(Data above based on benchtest at T-MOTOR lab are for reference only.)

Type	Propeller	Voltage (V)	Throttle (%)	Current (A)	RPM	Thrust (g)	Power (W)	Efficiency (g/W)	Torque (N*m)	Operating Temperature (℃)
F50 2150KV	HQ MCK	24.03	10%	0.38	5425	37	9	3.96	0.00	132
		24.01	20%	1.39	10223	144	33	4.33	0.02	
		23.99	30%	3.38	14796	316	81	3.90	0.03	
		23.94	40%	6.51	18827	528	156	3.39	0.05	
		23.88	50%	10.70	22363	758	256	2.97	0.08	
		23.80	60%	16.71	25841	1019	398	2.56	0.11	
		23.69	70%	23.89	28707	1281	566	2.26	0.13	
		23.57	80%	32.06	31353	1531	756	2.03	0.16	
		23.44	90%	41.15	33873	1788	964	1.85	0.19	
		23.29	100%	51.16	35724	1983	1192	1.66	0.21	

Note: Motor temperature is motor surface temperature running at 80% throttle for 10 seconds.
(Data above based on benchtest at T-MOTOR lab are for reference only.)

Type	Propeller	Voltage (V)	Throttle (%)	Current (A)	RPM	Thrust (g)	Power (W)	Efficiency (g/W)	Torque (N*m)	Operating Temperature (℃)
F50 2150KV	GF51377-3	24.03	10%	0.39	5294	40	9	4.29	0.00	122
		24.01	20%	1.48	10018	161	35	4.55	0.02	
		23.98	30%	3.69	14490	354	89	4.00	0.04	
		23.93	40%	7.02	18241	577	168	3.43	0.06	
		23.86	50%	11.63	21673	824	277	2.97	0.09	
		23.77	60%	18.29	25020	1125	435	2.59	0.12	
		23.64	70%	26.38	27739	1448	624	2.32	0.15	
		23.52	80%	34.84	30073	1715	820	2.09	0.18	

The manufacturer recommends to use this motor with active cooling, like a fan and heatsink. This is

		because it heats up incredibly quickly. I have chosen to use this motor as it is strong enough to produce the torque needed to generate consistent airflow, with a gearbox.. The motor itself is relatively inexpensive, accessible enough for now. The motor is also used for UAV's to drive propellers to incredibly high RPM, making it suitable for this application with a reduction.	
17.		This gearbox will be used in order to reduce the RPM from the motor and increase torque. This gearbox is manufactured and utilises metal gears, which makes it much less likely to fail as compared to 3d printed gears.	42mm Planetary Gearbox For 775/795/4260 Motor Reducer Large Torque 4/5mm Metal Motor Gear 8/10MM Shaft - AliExpress 1420 10/03/2026

18.		I will likely use tinkercad codeblocks in order to mass design the pipes as it would be incredibly tedious to design 48 individual pipes by hand. Tinkercad codeblocks would allow me to put in the parameters for the pipe geometry and then spit out any pipe I desire. This will only be used for the 3D printed pipes.	

19.		Tuning. In order for the pipes to be in tune with A=440Hz I will need to account for end correction- this means for a cylindrical bore pipe I will need to add 0.6 times the radius of the bore to the length of the pipe, otherwise it will sound flat.	

Pipes

Pipe Information

Note	Frequency	Length	Pipe type	Voicing	Length	Pipe type	Voicing
A4	440	389.77	Open	Tuning	190.25681	Closed	Tuning
C6	1046.5	163.88	Open	Principle	81.94	Closed	Flute
C#6	1108.7	154.68	Open	Principle	77.341	Closed	Flute
D6	1174.7	146	Open	Principle	73	Closed	Flute
D#6	1244.5	137.81	Open	Principle	68.903	Closed	Flute
E6	1318.5	130.07	Open	Principle	65.036	Closed	Flute
F6	1396.9	122.77	Open	Principle	61.385	Closed	Flute
F#6	1480	115.88	Open	Principle	57.94	Closed	Flute
G6	1568	109.38	Open	Principle	54.688	Closed	Flute

G#6	1661.2	103.24	Open	Principl e	51.619	Closed	Flute
A6	1760	97.443	Open	Principl e	48.722	Closed	Flute
Bb6	1864.7	91.974	Open	Principl e	45.987	Closed	Flute
B6	1975.3	86.821	Open	Principl e	43.41	Closed	Flute
C7	2093	81.94	Open	Principl e	40.97	Closed	Flute
C#7	2217.5	77.341	Open	Principl e	38.67	Closed	Flute
D7	2349.3	73	Open	Principl e	36.5	Closed	Flute
D#7	2489	68.903	Open	Principl e	34.451	Closed	Flute
E7	2637	65.036	Open	Principl e	32.518	Closed	Flute
F7	2793.8	61.385	Open	Principl e	30.693	Closed	Flute
F#7	2960	57.94	Open	Principl e	28.97	Closed	Flute
G7	3136	54.688	Open	Principl e	27.344	Closed	Flute
G#7	3322.4	51.619	Open	Principl e	25.809	Closed	Flute
A7	3520	48.722	Open	Principl e	24.361	Closed	Flute

Bb7	3729.3	45.987	Open	Principle	22.994	Closed	Flute
B7	3951.1	43.406	Open	Principle	21.703	Closed	Flute
C8	4186	40.97	Open	Principle	20.485	Closed	Flute

Tab 4

Github Logs:

This section contains an overview of my Github repository commits and releases.

OpenOrgel Synthesiser Pre-Alpha V.1.0 Pre-release 11:10, 03/06/2026

This is the distribution file for the first testing version of the OpenOrgel synthesis. New functionality TBA.

- Multiple stops across common footages, as well as mixture
- Compatability with Arduino Uno or Teensy 4.0 run organ console- currently only allowing for 8 switches due to physical limitations at this time. V2.0 will have full support of all current stops.
- Pitch circle
- Load MIDI

- Stop MIDI
- Pause MIDI
- Resume MIDI
- Realtime stop changing
- Motor control

Support for physical pipes via servos is to be implemented in Alpha phase.

OpenOrgel Synthesiser Pre-Alpha V.1.1 Pre-release

11:18, 03/06/2026

This is mainly just an optimisation update,

adding changes in source code into dist.

Should be marginally faster.

This release was made to fix my biggest issue with my synthesis engine right now- the lag. It takes forever to synthesise a single song. This adds caching which speeds it up greatly, it also made the memory allocations quite a lot better for the operating system to do, allowing for better compatibility and performance on a range of systems.